

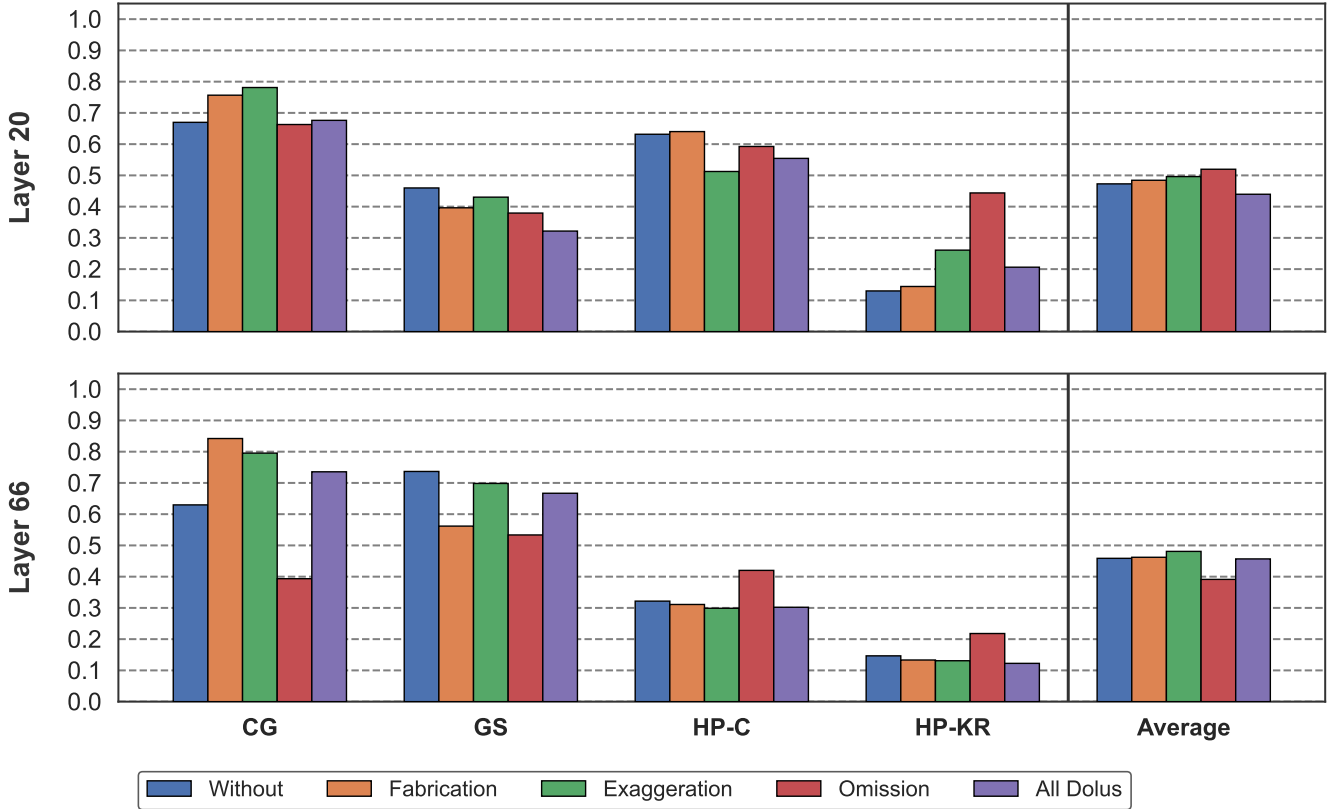


Figure 2: Comparison of logistic-probe AUROC across lie-typology training conditions and evaluation datasets. We compare probes trained without DolusChat, on fabrication-only DolusChat, on exaggeration-only DolusChat, on omission-only DolusChat, and on the full DolusChat mixture, at 20th-percentile and 66th-percentile.

Crucially, while INLP serves as a useful geometric baseline, its performance should be interpreted as a measure of subspace robustness rather than standard detection. As shown in Table 1, INLP is often close to logistic regression, but not identical across all datasets and depths: on CG it reaches 0.74 and 0.82 AUROC versus 0.76 and 0.84 for logistic regression, on GS it reaches 0.41 and 0.55 versus 0.40 and 0.56, on HP-C it matches the baseline at 0.64 and 0.31, and on HP-KR both methods remain near chance. These results suggest that iteratively projecting out linear directions removes some information, but does not uniformly collapse the deception signal. Rather than indicating that deception is confined to a single low-dimensional subspace, the findings are more consistent with a partly distributed representation that is only partially removed by linear null-space projection.

5.3 Impact of Sparse SAE Features

Unlike the earlier fabrication-only comparison, this experiment uses SAE results computed across all DolusChat lie categories. Furthermore, due to computational constraints, SAE features were only extracted at 66th percentile layer. To keep the comparison matched, we therefore evaluate the dense hidden on the same all-category DolusChat subset and at the same layer, so any differences reflect

the representation type rather than changes in lie-type coverage or depth.

To examine whether sparse SAE features preserve or improve deception separability relative to dense hidden states, we compare Gemma-2-27B hidden activations against Gemma-2-27B SAE features at 66th percentile layer in Table 2.

The results presented in the table suggest that, within this specific experimental setup, the transition from dense hidden states to sparse SAE features does not appear to yield a clear improvement in deception detection performance. While the sparse representations maintain a relative functional parity with the dense branch, they generally achieve slightly lower or equivalent average scores across most datasets and probing methods. This suggests that the SAE reconstruction at this depth preserves existing deception-related signals rather than enhancing their separability.

The main exception is the harm-pressure setting, where SAE features improve some individual probe scores and appear to help disentangle deception from overlapping safety-related signals. However, this effect is not uniform across all methods or datasets, and it does not translate into a consistent overall gain. The upper-bound rows further support this interpretation: dense and sparse features

remain close in performance, but the dense branch is still slightly stronger overall.

5.4 Impact of Lie Typology

To test whether the structural category of a lie affects detectability, we compare AUROC across five training conditions: without DolusChat, fabrication-only, exaggeration-only, omission-only, and the full DolusChat mixture. We restrict this analysis to the baseline logistic regression probe because, as established in Section 5.2, it provides the most stable and representative baseline across datasets. By standardizing on this robust linear boundary, we successfully isolate the effect of the training distribution on linear separability without compounding variables from fluctuating probe expressivity. Figure 2 summarizes these results at the 20th percentile layer and the 66th percentile layer.

The bar plot shows that lie typology matters, but not in a simple universal order. At the 20th percentile, the best-performing condition depends on the dataset: exaggeration is strongest on CG, the no-DolusChat baseline is strongest on GS, fabrication is strongest on HP-C, and omission is strongest on HP-KR. Averaged across datasets, omission has the highest AUROC at the 20th percentile layer, largely because of its strong performance on HP-KR. This suggests that omission is the most recoverable training condition at the shallower layer, but only in the harm-pressure setting.

At the 66th percentile layer, the pattern shifts. Fabrication becomes strongest on CG, the no-DolusChat baseline remains strongest on GS, and omission becomes the strongest condition on both HP-C and HP-KR. On average, exaggeration achieves the highest AUROC at the 66th percentile layer, while omission drops substantially relative to its 20th percentile layer performance. This indicates that deeper representations sharpen some lie-typology signals while suppressing others, especially when the lie depends on contextual withholding rather than direct contradiction.

The full DolusChat mixture is never the best-performing condition and is weakest at the 20th percentile layer, with a near-bottom average at the 66th percentile layer as well. This suggests that combining lie types does not automatically improve linear separability and may instead blur the deception signal. Overall, the figure supports a qualified conclusion: lie typology does affect detectability, but there is no single ordering of fabrication, exaggeration, and omission that holds across all datasets or depths.

Summary: A Representation-Dependent Problem. The results point to a consistent but nuanced picture of deception detection in LLMs. Depth improves separability in some datasets but not others, so later layers are not universally superior. More expressive probes help selectively, but standard logistic regression remains highly competitive and is often the most stable baseline. Sparse SAE features perform similarly to dense representations. Finally, lie typology clearly matters: fabrication, omission, exaggeration, and mixed training conditions produce different detection patterns depending on the dataset and representation layer. Overall, the experiments show that deception detection is best understood as a representation-dependent problem rather than a single-classifier problem.

6 Conclusion

This paper set out to examine deception detection in large language models through four factors: representation depth, probe expressivity, sparse representations, and lie typology. Across the experiments, the central conclusion is that deception is not encoded in a single uniform way. Instead, detectability depends heavily on the dataset context, the layer being probed, and the representation space used for classification.

For depth, the results show a dataset-dependent pattern rather than a universal rule. Some datasets benefit from deeper layers, while others remain more separable at earlier layers, meaning the commonly assumed “later is better” trajectory does not hold consistently across all deception settings. For probe expressivity, more complex models provide selective gains but do not dominate the simple logistic baseline. Truth2D offers the clearest structured improvement, while TPC, mass-mean, and INLP are useful in specific settings but not consistently stronger than a well-calibrated linear classifier. This suggests that deception detection is not solved simply by adding geometric or polynomial complexity.

For sparse SAE features, the results suggest that evaluating SAE representations at 66th percentile layer does not consistently enhance detectability over the original dense hidden states. In most settings, SAE preserves the core deceptive signal rather than improving it, and its benefits appear limited to a subset of harm-pressure cases. This suggests that, at least at this depth, SAE features function more as an alternative representation than as a reliably stronger one for deception detection. Finally, for lie typology, we found that fabrication, omission, exaggeration, and mixed training conditions do not behave uniformly. The strongest condition depends on both the dataset and the representation layer, indicating that the structural form of a lie materially affects its detectability, and no single lie type dominates universally.

The main challenge throughout this paper was that the deception signal was not stable across tasks; different datasets exposed different failure modes, and the optimal probing strategy frequently shifted with the evaluation context. Given that optimal detection performance depends heavily on highly localized factors—such as specific layers, lie typologies, and representation spaces—our findings strongly suggest that a monolithic detector is fundamentally limited. Instead, a highly promising avenue for future work lies in ensemble methods or a Mixture-of-Experts (MoE) probing framework. By training specialized, expert probes for distinct context types (e.g., separating safety-related choices from persona-driven roleplay) and utilizing a gating mechanism to dynamically route activations, such an architecture could explicitly leverage the selective strengths of the individual probes observed in our evaluation. Additionally, extending this analysis to the remaining Liars’ Bench models would test whether these depth, sparsity, and typology patterns persist across different model families, while studying the SAE representations more directly could further bridge the gap between behavioral benchmarking and mechanistic interpretability.

Limitations

This study has several limitations. First, given the computational scope of a multi-factor study and the constraints of an extended abstract, we sampled two specific depth percentiles (20% and 66%)

rather than conducting a continuous layer-wise sweep. Because we purposefully anchored our early layer to the original Liars' Bench baseline and our deeper layer to the maximum theoretical peak before the late-stage performance drop-off identified by Boxo et al. [3], we chose to apply these established bounds directly to our evaluation rather than exhaustively re-proving a continuous layer-wise trajectory from scratch. Second, because our core claims are intentionally conservative—highlighting contextual dependencies (“it depends”) rather than declaring absolute algorithmic superiority—we rely on directional shifts in AUROC and refrain from exhaustive statistical significance testing. Third, the lie typology experiments (Figure 2) were restricted to the logistic regression baseline to cleanly isolate distribution effects, and have not yet been evaluated across the more expressive probe families. Fourth, the SAE results were evaluated solely at 66th percentile layer. Fifth, the inherent distribution gap between our synthetic training data (DolusChat) and the complex evaluation prompts (Liars' Bench) likely contributes to the striking anti-transfer observed in the HP datasets. Finally, our discrete classification metrics rely on a default 0.5 probability threshold, whereas future operational work must explore strict False Positive Rate (FPR) constraints.

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